Input-files of changeResistivity

Input-files of changeResistivity

File name	Contents
Arbitrary name	Controlling parameters
mesh.dat	Mesh information
resistivity_block_iterX.dat *	Information of parameter cells

* In the file name, ‘X ‘ indicates the iteration number.

How to use changeResistivity

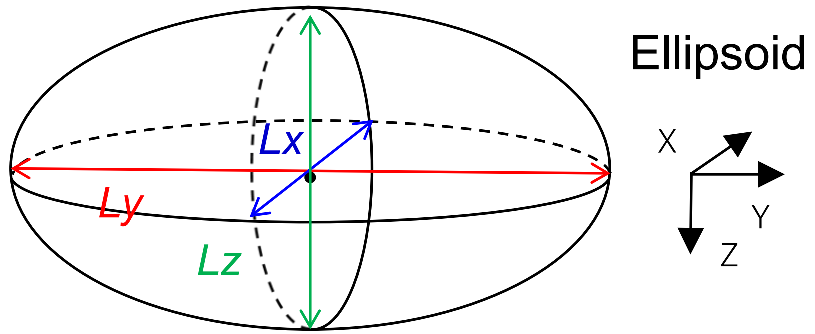
You need to execute the following command in the directory where input files exist.

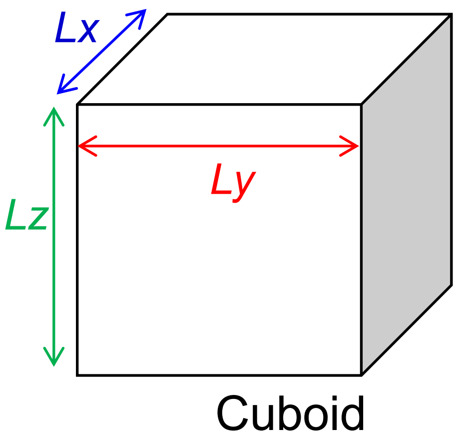
```
changeResistivity     [Name of parameter file]     [-aniso]
```

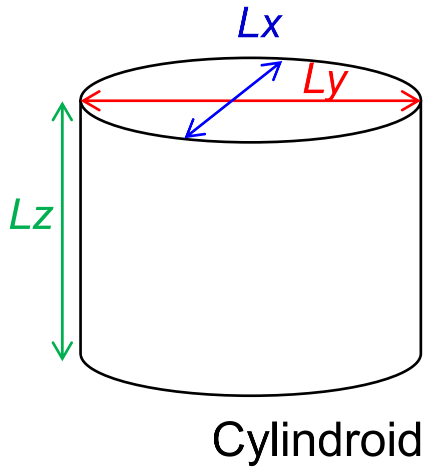
The optional argument ‘-aniso’ is required when the file for the anisotropic inversion is used.

File format of parameter file for the isotropic case (by default)

Iteration number
Shape of the area in which resistivity values are changed [1)]
Lx (km) · *Ly (km)* · *Lz (km)* — *Lengths along each axis*
X (km) · *Y (km)* · *Z (km)* — *Center of the area*
Rotation angle of the area around the z-axis (deg.)
Lower limit of the resistivity values changed in this program(Ωm) [2)]
Upper limit of the resistivity values changed in this program (Ωm) [3)]
Resistivity after the change (Ωm)
Lower limit of the resistivity after the change (Ωm)
Upper limit of the resistivity after the change (Ωm)







1) 0 ➡ Ellipsoid, 1 ➡ Cuboid, 2 ➡ Cylindroid

2) If the resistivity value is lower than the lower limit, the resistivity value is not changed.

3) If the resistivity value is higher than the upper limit, the resistivity value is not changed.

File format of parameter file for the anisotropic case (when the argument '-aniso' is used)

Iteration number

Shape of the area in which resistivity values are changed

Lx (km)	*Ly (km)*	*Lz (km)*	*Lengths along each axis*
X (km)	*Y (km)*	*Z (km)*	*Center of the area*

Rotation angle of the area around the z-axis (deg.)

Lower limit of ρ_{xx} *to be changed* (Ωm)	*Upper limit of* ρ_{xx} *to be changed* (Ωm)	ρ_{xx} *after the change* (Ωm)
Lower limit of ρ_{yy} *to be changed* (Ωm)	*Upper limit of* ρ_{yy} *to be changed* (Ωm)	ρ_{yy} *after the change* (Ωm)
Lower limit of ρ_{zz} *to be changed* (Ωm)	*Upper limit of* ρ_{zz} *to be changed* (Ωm)	ρ_{zz} *after the change* (Ωm)
Lower limit of θ_S *to be changed* (deg.)	*Upper limit of* θ_S *to be changed* (deg.)	θ_S *after the change* (deg.)
Lower limit of θ_D *to be changed* (deg.)	*Upper limit of* θ_D *to be changed* (deg.)	θ_D *after the change* (deg.)
Lower limit of θ_L *to be changed* (deg.)	*Upper limit of* θ_L *to be changed* (deg.)	θ_L *after the change* (deg.)

Output-files of changeResistivity

*'X ' indicates the iteration number.

Output files for the isotropic case (by default)

File name	Contents
resistivity_block_iterX.mod.dat *	Information of parameter cells after the change of resistivity
ResistivityMod.iterX	Resistivity values of each element (Ensight Gold file format (binary))

Output files for the anisotropic case (when the argument '-aniso' is used)

File name	Contents
resistivity_block_iterX.mod.dat *	Information of parameter cells after the change of resistivity
AnisotropyTypesMod	Anisotropy type (0: isotropy, 1: transverse isotropy, 2: general anisotropy)
RhoXXMod.iterX *	Value of ρ_{xx} (Ωm) (EnSight Gold variable file format)
RhoYYMod.iterX *	Value of ρ_{yy} (Ωm) (EnSight Gold variable file format)
RhoZZMod.iterX *	Value of ρ_{zz} (Ωm) (EnSight Gold variable file format)
StrikeMod.iterX *	Value of θ_S (deg.) (EnSight Gold variable file format)
DipMod.iterX *	Value of θ_D (deg.) (EnSight Gold variable file format)
SlantMod.iterX *	Value of θ_L (deg.) (EnSight Gold variable file format)
AnisotropyMod.iterX *	$\lvert \log(\rho_{xx}) - \log(\rho_{yy}) \rvert$ (EnSight Gold variable file format)